ARTIFICIAL INTELLIGENCE (CT 653)

Tribhuvan University | Institute of Engineering | BCT | Year III / Part II

CHAPTER 5: Structured Knowledge Representation

Syllabus Hours: 4 hrs | Marks Weight: 7 marks

Complete Exam-Ready Answers | All Past Questions Covered

CHAPTER 5 SYLLABUS TOPICS

- 5.1 Representations and Mappings
- 5.2 Approaches to Knowledge Representation
- 5.3 Issues in Knowledge Representation
- 5.4 Semantic nets, Frames
- 5.5 Conceptual dependencies and Scripts

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SECTION 1: Representations, Mappings & Approaches to Knowledge Representation

Syllabus Ref: 5.1, 5.2, 5.3 | Exam Ref: 2080 Chaitra, 2081 Ashwin, 2082 Kartik, 2079 Ashwin Back, 2078 Chaitra

Q1. What is knowledge representation? What are the different approaches to knowledge representation? How do you make a choice among these approaches? Is it possible to evaluate the knowledge representation? List down any issues associated with it. **[2+2+2+2]** (2080 Chaitra)

Answer: Knowledge Representation

Knowledge Representation (KR) is the method by which information about the world is encoded in a computer system so that an AI agent can use it to solve complex problems, make decisions, and draw inferences.

A good knowledge representation system must be able to:

- Represent all the knowledge needed in a given domain.
- Be used by the AI agent to reason and derive new knowledge.
- Be understood and modified by humans (knowledge engineers).
- Support efficient retrieval and inference.

Representations and Mappings: Knowledge representation involves creating a MAPPING between the real world and a formal, symbolic internal representation. There are two levels:

Level	Description	Example
World Level	Facts and relationships as they exist in reality	Socrates was a human philosopher
Symbol Level	Formal encoding of those facts inside the computer	Human(Socrates), Philosopher(Socrates)

The mapping must be consistent, complete, and unambiguous for the AI system to reason correctly.

Approaches to Knowledge Representation:

Approach	Description	Strengths	Weaknesses
Logical Representation (Propositional & FOPL)	Uses formal logic with symbols and rules	Precise, provable, supports inference	Verbose, computationally expensive
Semantic Networks	Graph with nodes (concepts) and edges (relations)	Intuitive, supports inheritance	Hard to represent complex relations
Frames	Record-like structures with slots and fillers	Organizes related info together, supports defaults	Rigid structure, inheritance issues
Production Rules (IF-THEN)	Set of condition-action rules	Modular, easy to add/remove rules	Slow for large rule sets, no learning
Scripts	Sequence of events for stereotyped situations	Good for common sense reasoning	Requires pre-defined scenarios

Conceptual Dependency (CD)	Represents actions in terms of primitives	Language-independent, captures meaning	Complex representation

How to Choose an Approach:

1. Nature of the domain: If the domain has clear logical rules → use FOPL. If it involves classification/hierarchy → use Semantic Networks or Frames.

2. Type of reasoning needed: For deductive reasoning → Logic. For pattern matching → Production Rules. For event-based reasoning → Scripts.

3. Efficiency requirements: For fast inference → Production Rules. For complex logical proofs → FOPL with Resolution.

4. Ease of modification: Production rules are easy to add/remove. Frames are easy to update for specific objects.

5. Inheritance needs: Semantic Networks and Frames naturally support inheritance of properties.

Evaluating Knowledge Representation:

Yes, it is possible to evaluate a KR system. Evaluation criteria include:

- Representational Adequacy: Can it represent all needed knowledge?
- Inferential Adequacy: Can it derive all needed conclusions?
- Inferential Efficiency: How quickly can it derive conclusions?
- Acquisitional Efficiency: How easily can new knowledge be added?
- Clarity and Understandability: Can humans easily understand the representation?

Issues in Knowledge Representation:

1. Knowledge Acquisition Bottleneck: Extracting knowledge from domain experts is slow, expensive, and error-prone.

2. Completeness: No representation can capture ALL knowledge about a complex domain.

3. Consistency: Conflicting facts may be present in large knowledge bases.

4. Ambiguity: Natural language is inherently ambiguous — hard to represent precisely.

5. Frame Problem: How to represent what changes and what stays the same after an action.

6. Uncertain Knowledge: Real-world knowledge is often incomplete or probabilistic.

7. Scalability: As the knowledge base grows, reasoning becomes slower.

Q2. What are the essential properties and issues of knowledge representation? What are the different approaches for knowledge representation? List down the essential properties. [4+2+2] (2079 Ashwin Back)

Answer: Essential Properties of a KR System

Property	Description
Representational Adequacy	Ability to represent all types of knowledge needed (facts, rules, relations, uncertainty)
Inferential Adequacy	Ability to derive new knowledge from existing knowledge through inference
Inferential Efficiency	Ability to derive conclusions quickly using heuristics and indexing

Acquisitional Efficiency	Ease with which new knowledge can be added without restructuring the entire KB
Modularity	Knowledge should be organized in modular, independent units
Transparency	The knowledge should be understandable to humans, not just machines
Consistency	No contradictory facts should exist in the knowledge base
Expressiveness	Ability to express complex relationships and constraints

The different approaches (Logical, Semantic Networks, Frames, Production Rules, Scripts, Conceptual Dependency) are detailed in Q1 above.

SECTION 2: Semantic Networks — Most Repeated Topic in Chapter 5

Exam Ref: 2082 Kartik, 2081 Ashwin, 2081 Chaitra, 2080 Ashwin Back, 2079 Chaitra, 2079 Ashwin Back, 2078 Chaitra, 2078 Poush, 2077 Chaitra, 2076 Baisakh, 2075 Bhadra, 2073 Bhadra, 2072 Magh, 2071 Bhadra, 2070 Magh

Q3. Define term Semantic Net. Represent the following sentences using Semantic Net representation. Tom is a cat. Tom caught a bird. Tom is owned by Sudeep. Tom is ginger in color. Cats like cream. The cat sat on the mat. A cat is a mammal. A bird is an animal. All mammals are animals. Mammals have fur. Sudeep age is 20. **[2+6]** (2081 Chaitra)

Answer: Semantic Network

A Semantic Network (Semantic Net) is a graphical knowledge representation technique where:

- NODES represent objects, concepts, or events.
- ARCS (edges) represent relationships between nodes.
- Common relations: ISA (is-a), AKO (a-kind-of), HAS (has-property), OWNS, CAUGHT, COLOR, LIVES-ON, AGE, LIKES

Semantic networks support INHERITANCE — if Cat ISA Mammal and Mammal HAS Fur, then Cat automatically inherits Fur.

Semantic Network for Given Sentences:

From Node	Relationship	To Node	Source Sentence
Tom	ISA	Cat	Tom is a cat
Tom	CAUGHT	Bird (instance)	Tom caught a bird
Tom	OWNED-BY	Sudeep	Tom is owned by Sudeep
Tom	COLOR	Ginger	Tom is ginger in color
Tom	SAT-ON	Mat	The cat sat on the mat
Cat	LIKES	Cream	Cats like cream
Cat	ISA	Mammal	A cat is a mammal
Bird	ISA	Animal	A bird is an animal
Mammal	ISA	Animal	All mammals are animals
Mammal	HAS	Fur	Mammals have fur
Sudeep	AGE	20	Sudeep age is 20

Textual Graph Representation:

```
                    ANIMAL
                  /        \
                ISA        ISA
                /            \
            MAMMAL          BIRD
           /    \             \
       ISA     HAS           ISA
       /         \             \
    CAT          Fur        Bird(inst)
   / | \                        ^
```

```
      ISA | LIKES            CAUGHT
      /   |   \                 |
    Tom  SAT  Cream            Tom
     |\   |
     | \  Mat
   COLOR OWNED-BY
     |       |
   Ginger  Sudeep
             |
            AGE
             |
            20
```

Key advantage: Tom automatically inherits properties like HAS Fur (from Mammal) and ISA Animal (through Cat → Mammal → Animal) without explicitly stating them — this is the power of inheritance in semantic networks.

Q4. What is semantic network? Convert the following sentences into semantic network. i) The avg height of an adult is 5'6". ii) Cricket player is an adult male. iii) Adult male is a person. iv) Batting average of a cricket player is 37.59. v) Rohit Paudel is a Fielder. vi) Fielder is a cricket player. vii) Team of Rohit Paudel is Nepal. [2+6] (2079 Chaitra)

Answer:

From Node	Relationship	To Node
Adult	AVG-HEIGHT	5'6"
Cricket Player	ISA	Adult Male
Adult Male	ISA	Person
Cricket Player	BATTING-AVG	37.59
Rohit Paudel	ISA	Fielder
Fielder	ISA	Cricket Player
Rohit Paudel	TEAM	Nepal

Inherited facts (through ISA chain):

- Rohit Paudel ISA Fielder ISA Cricket Player → Rohit Paudel's batting average is 37.59 (inherited).
- Rohit Paudel ISA Fielder ISA Cricket Player ISA Adult Male ISA Person → Rohit Paudel is a Person (inherited).
- Rohit Paudel's avg height = 5'6" (inherited from Adult through the ISA chain).

Q5. What are Frames and Semantic Net? Convert the following sentences into Semantic Net: A person is a mammal. Sandeep Lamichane is a person. Person has nose. Sandeep Lamichane is in Nepalese cricket team. Uniform color of Sakti Gauchan is Red/Blue. [2+6] (2078 Poush, 2072 Magh)

Answer: Frames vs Semantic Networks

Feature	Semantic Network	Frame
Structure	Graph (nodes and edges)	Record/table (slots and fillers)
Representation	Visual, intuitive relationships	Structured, attribute-value pairs
Inheritance	Through ISA/AKO links	Through class hierarchy
Best for	Showing relationships and hierarchy	Describing a single entity in detail
Flexibility	Less structured	More structured and organized

Semantic Network for Given Sentences:

From Node	Relationship	To Node
Person	ISA	Mammal
Sandeep Lamichane	ISA	Person
Person	HAS	Nose
Sandeep Lamichane	MEMBER-OF	Nepalese Cricket Team
Sakti Gauchan	UNIFORM-COLOR	Red/Blue

```
MAMMAL
  |
 ISA
  |
PERSON ---- HAS ----> Nose
  |
 ISA
  |
Sandeep Lamichane ---- MEMBER-OF ----> Nepalese Cricket Team

Sakti Gauchan ---- UNIFORM-COLOR ----> Red/Blue
```

Inherited: Sandeep Lamichane HAS Nose (inherited from Person). Sandeep Lamichane ISA Mammal (through Person).

Q6. What are Frames and Semantic Network? Convert the following sentences into FOPL and represent using semantic network. i) Ram is a person. ii) Person is an animal. iii) Ram likes Jhilke. iv) Jhilke is a cat. v) Cat eats fish. vi) Cat is an animal. vii) Fish is an animal. [2+6] (2081 Ashwin)

Answer: FOPL Representation:

1. Person(Ram)
2. $\forall x$ Person(x) $\rightarrow$ Animal(x)
3. Likes(Ram, Jhilke)
4. Cat(Jhilke)

5. ∀x Cat(x) → Eats(x, Fish)
6. ∀x Cat(x) → Animal(x)
7. Animal(Fish)

Semantic Network:

From Node	Relationship	To Node
Ram	ISA	Person
Person	ISA	Animal
Ram	LIKES	Jhilke
Jhilke	ISA	Cat
Cat	EATS	Fish
Cat	ISA	Animal
Fish	ISA	Animal

```
          ANIMAL
        /   |   \
      ISA  ISA   ISA
      /     |     \
  PERSON  CAT   FISH
    |    /   \    ^
   ISA  ISA  EATS|
    |   |        |
   Ram Jhilke----+
     \
     LIKES
       \
      Jhilke
```

SECTION 3: Frames — Detailed Theory and Examples

Exam Ref: 2082 Kartik, 2081 Ashwin, 2080 Chaitra, 2079 Ashwin Back, 2078 Chaitra, 2075 Baisakh, 2074 Bhadra

Q7. What are different issues in knowledge representation? Discuss the importance of Frame in knowledge representation with suitable example. [3+5] (2080 Ashwin Back)

Answer: Frames in Knowledge Representation

A FRAME is a data structure for representing a stereotyped situation or object. Introduced by Marvin Minsky (1975), a frame contains:

- SLOT NAME: The attribute being described (e.g., Color, Size, Owner).
- FILLER: The value of the slot (e.g., Red, Large, John).
- DEFAULT VALUE: A value assumed if no specific value is given.
- CONSTRAINTS: Restrictions on acceptable filler values.
- PROCEDURAL ATTACHMENT: A procedure (demon) that runs when the slot is accessed (IF-NEEDED, IF-ADDED).

Importance of Frames:

1. Organized Representation: All information about an entity is stored in one place.
2. Default Reasoning: If a slot has no value, the default is used — useful for incomplete information.
3. Inheritance: Subframes inherit slot values from parent frames (e.g., a 'Robin' frame inherits from 'Bird' frame).
4. Modularity: Each frame is independent and can be modified without affecting others.
5. Prototypical Knowledge: Frames capture typical/default knowledge about categories.

Example: Frame for a Car

Frame: CAR	
ISA	Vehicle
Number-of-Wheels	4 (default)
Engine-Type	Petrol/Diesel/Electric
Color	Unknown (to be filled)
Max-Speed	120 km/h (default)
Owner	Unknown
IF-NEEDED (Price)	Call procedure: lookup_price()

Example: Frame Hierarchy for Animals

Frame: ANIMAL	
ISA	Living-Being
Breathes	Yes (default)
Moves	Yes (default)
Reproduces	Yes

Frame: MAMMAL (subframe of ANIMAL)	
ISA	Animal
Warm-Blooded	Yes (default)
Has-Hair	Yes (default)
Birth-Type	Live-birth (default)

Frame: DOG (subframe of MAMMAL)	
ISA	Mammal
Sound	Bark
Legs	4
Domesticated	Usually-Yes
Famous-Example	Lassie

Inheritance in action: A DOG frame inherits: Breathes=Yes, Moves=Yes, Warm-Blooded=Yes, Has-Hair=Yes from parent frames, without explicitly stating them.

Q8. What is Frame? How is it different from semantic net in knowledge representation? [7]
(2074 Bhadra)

Answer: Frame vs Semantic Network — Detailed Comparison

Aspect	Frame	Semantic Network
Definition	Record-like structure with slots (attributes) and fillers (values)	Graph structure with nodes (concepts) and labeled arcs (relations)
Representation	Tabular/record format — groups all info about one entity	Graph format — shows relationships between many entities
Inheritance	Explicit through class-subclass hierarchy of frames	Implicit through ISA / AKO links in graph
Default Values	Supported — slots can have default values	Not directly supported
Procedural Attachment	Supported via IF-NEEDED / IF-ADDED demons	Not directly supported
Modification	Easy to add/modify slots for a specific frame	Must modify graph nodes and edges
Expressiveness	Richer — supports constraints, procedures, defaults	Simpler — primarily relational
Visual Clarity	Less visual, more structured	Highly visual, intuitive

Best Used For	Describing a single complex entity in detail	Showing relationships across many concepts
Example	Frame: Student [Name: Ram, Age: 20, Dept: BCT]	Ram ISA Student; Student ISA Person

SECTION 4: Conceptual Dependency & Scripts

Syllabus Ref: 5.5 | Exam Ref: 2077 Chaitra, 2071 Bhadra, 2070 Magh

Q9. What do you mean by Conceptual Dependency? Explain some of the common primitives used in conceptual dependency. [2+5] (2077 Chaitra, 2071 Bhadra)

Answer: Conceptual Dependency (CD)

Conceptual Dependency (CD) is a theory of knowledge representation developed by Roger Schank (1973). It represents the MEANING of natural language sentences in a language-independent, canonical form using a set of primitive actions.

Key Idea: Two sentences that have the same meaning should have the SAME CD representation, regardless of the language or phrasing used.

Example:

- 'John gave Mary a book' and 'Mary received a book from John' → same CD representation.

Basic Structure:

```
Actor ⟷ ACT ⟷ Object
            |
        Direction (to/from)
            |
      Instrumental (how)
```

Primitive Acts (ACTs) in Conceptual Dependency:

Primitive	Meaning	Example Sentence	CD Notation
ATRANS	Transfer of abstract possession (ownership, control)	John gave Mary a book.	John ATRANS book to Mary from John
PTRANS	Physical transfer of location of an object	John went to school.	John PTRANS John to school from home
MTRANS	Transfer of mental information (telling, reading)	John told Mary the news.	John MTRANS news to Mary's mind
MBUILD	Building new mental information (thinking, deciding)	John decided to leave.	John MBUILD (leave) in John's mind
PROPEL	Application of physical force to an object	John pushed the door.	John PROPEL door direction-forward
MOVE	Movement of body part by owner	John raised his hand.	John MOVE hand direction-up
INGEST	Taking a substance into the body	John ate the pizza.	John INGEST pizza into John's mouth
EXPEL	Expelling a substance from the body	John cried.	John EXPEL tears from John's eyes
SPEAK	Producing a sound	John said 'hello'.	John SPEAK 'hello' to Mary

ATTEND	Focusing a sense organ on a stimulus	John looked at Mary.	John ATTEND eyes to Mary
GRASP	Grasping an object physically	John held the pen.	John GRASP pen
DO	A general action (when no specific primitive fits)	John exercised.	John DO exercise

Full Example — 'John gave Mary a book':

```
John ←ACT: ATRANS→ Book
          TO: Mary
          FROM: John
[Meaning: The abstract possession/control of the book transferred from John to Mary]
```

Q10. What is Conceptual Dependency? Explain how knowledge is represented using Scripts. [3+5] (2077 Chaitra)

Answer: Scripts

A SCRIPT (developed by Roger Schank and Robert Abelson, 1977) is a structured representation that describes a SEQUENCE OF EVENTS that typically occur in a stereotyped situation (e.g., going to a restaurant, visiting a doctor, buying a ticket).

Components of a Script:

Component	Description	Example (Restaurant Script)
Entry Conditions	Conditions that must be true before the script is applicable	Customer is hungry. Restaurant is open.
Roles	The participants involved in the script	Customer, Waiter, Cook, Cashier
Props	Objects used in the script	Menu, Table, Food, Bill, Money
Scenes	The sequence of sub-events (tracks)	Entering, Ordering, Eating, Leaving
Results	Facts that become true after the script completes	Customer is full. Restaurant has money.

Full Restaurant Script Example:

Script Name: RESTAURANT

Entry Conditions: Customer is hungry. Customer has money.

Roles: Customer (C), Waiter (W), Cook (K), Cashier (Ca)

Props: Menu (M), Table (T), Food (F), Bill (B), Money ($)

Results: Customer is not hungry. Customer has less money. Restaurant has more money.

Scenes:

- Scene 1 — ENTERING: C walks in. C looks for table. W leads C to table. C sits down.
- Scene 2 — ORDERING: W brings menu. C reads menu. C decides what to order. C signals W. W comes. C tells W order. W writes order. W gives order to K. K prepares food.
- Scene 3 — EATING: W brings food to C. C eats food. C finishes eating.
- Scene 4 — LEAVING: W brings bill. C reads bill. C gives money to W or Ca. Ca gives change. C leaves tip. C leaves restaurant.

How Scripts Aid Reasoning:

1. Scripts fill in missing details — if a story says 'John went to a restaurant and left a tip', the system knows John also ordered food, ate it, received a bill, etc.

2. Scripts allow the system to understand what is 'expected' in a situation and detect when something unusual happens.

3. Scripts enable natural language understanding by providing context for ambiguous sentences.

SECTION 5: Comparison & Combined Questions

Exam Ref: 2078 Baishakh, 2075 Bhadra, 2074 Bhadra, 2072 Ashwin, 2071 Bhadra

Q11. Explain frame and Semantic Net with example. List down their advantages and disadvantages. [4+4] (2076 Baisakh, 2073 Magh)

Answer: Advantages and Disadvantages

Semantic Networks:

Advantages	Disadvantages
Intuitive and easy to visualize	Hard to represent complex propositions (e.g., negation, disjunction)
Supports inheritance of properties through ISA/AKO links	No standard notation — different systems use different conventions
Efficient for querying relationships	Difficult to represent procedural knowledge
Natural way to represent taxonomic knowledge	Searching large networks can be slow
Good for organizing hierarchical knowledge	Ambiguous — same concept may be represented differently by different designers

Frames:

Advantages	Disadvantages
Highly structured — all info about entity in one place	Rigid structure — may not suit all domains
Supports default values for incomplete information	Inheritance conflicts can arise (diamond problem)
Procedural attachment allows active knowledge	Not easy to represent dynamic/changing situations
Easy to implement and understand	Knowledge acquisition for large frame systems is complex
Natural for object-oriented domains	Hard to represent general/universal rules

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Q12. What is knowledge representation? How semantic network is used to represent knowledge? [2+6] (2075 Bhadra)

Answer: How Semantic Networks Represent Knowledge

Semantic networks represent knowledge through a directed graph where:

1. IS-A relationships represent class membership (instance): 'Lassie IS-A Dog' — Lassie is an instance of Dog.
2. AKO (A-Kind-Of) relationships represent class-subclass: 'Dog AKO Animal' — Dog is a subclass of Animal.
3. HAS relationships represent properties: 'Dog HAS Tail', 'Dog HAS 4-Legs'.
4. Verb relations represent actions/events: 'John LOVES Mary', 'Cat EATS Mouse'.

5. Attribute relations: 'Rose COLOR Red', 'Elephant SIZE Large'.

Inheritance in Semantic Networks: If Dog AKO Animal and Animal HAS Lungs, then Dog HAS Lungs (inherited). This allows compact representation without repeating facts for every subclass.

Example:

Fact	Semantic Network Arc
Lassie is a dog	Lassie IS-A Dog
Dog is an animal	Dog AKO Animal
Animals breathe	Animal HAS Breathes=Yes
Lassie is brown	Lassie COLOR Brown
Lassie can bark	Lassie CAN Bark
Lassie's owner is John	Lassie OWNED-BY John

Reasoning Example: Question: Can Lassie breathe?

- Check: Lassie IS-A Dog
- Check: Dog AKO Animal
- Check: Animal HAS Breathes=Yes
- Conclusion: Yes, Lassie can breathe (through inheritance chain).

QUICK REVISION: Chapter 5 — Key Points Summary

Topic	Key Points to Remember	Expected Marks
Knowledge Representation	Mapping between world and symbolic level; 4 essential properties	2-4
Approaches to KR	6 approaches: Logic, Semantic Net, Frames, Production Rules, Scripts, CD	4-6
Issues in KR	Acquisition bottleneck, completeness, consistency, frame problem, ambiguity	2-4
Semantic Networks	Nodes=concepts, Edges=relations (ISA, AKO, HAS); supports inheritance	6-8
Frames	Slots + Fillers + Defaults + Procedural attachment; hierarchy with inheritance	4-6
Conceptual Dependency	12 primitive acts: ATRANS, PTRANS, MTRANS, MBUILD, PROPEL, INGEST...	4-6
Scripts	5 parts: Entry Conditions, Roles, Props, Scenes, Results; Restaurant example	4-6

Best of Luck! | TU IOE | BCT | CT 653 Artificial Intelligence | Chapter 5